THE UNIVERSITY OF AUCKLAND

SEMESTER ONE 2024
Campus: City

COMPUTER SCIENCE

AI Architecture and Design

(Time Allowed: TWO hours)

NOTE:

- This exam is out of **80** marks. Attempt **ALL** questions.
- Write your answers in the separate booklet provided to you.
- This is a restricted book test. You are only allowed to use a **double-sided page of notes**.
- Digital devices, including calculators and phones are not permitted.
- Cross out notes and other work you do not wish to be assessed.
- We privilege quality over quantity, i.e., you do not need to write very long answers. Be concise and clear.
- Write as clearly as possible, in plain English.

Answer ALL questions.

Part 1: Data and Deep Neural Networks [27 marks]

Question 1: Unbalanced data [8 marks]

You are working on a binary classification task for fraudulent transaction detection. The goal is to predict if a transaction is fraudulent or not. When exploring your data, you discover that there is imbalance in the data, with the number of fraudulent transaction samples representing 5% of the data set. Explain when and why this is can be problem, and briefly describe 2 different approaches you could use to try and alleviate this imbalance when training a model.

Question 2: Theorem of the Universal Function Approximators [4 marks]

The theorem of the universal function approximators states that a two-layer neural network (one hidden layer) with a sufficient number of neurons can approximate any continuous function to any desired accuracy. Explain in your own words why this is the case. You can use sketches if you like.

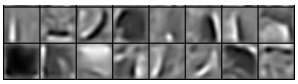
Question 3: Design choices [9 marks]

You trained a simple neural network with the following settings: 2 hidden layers, 5 neurons per layer with ReLU activation, random initialisation, trained for 100 epochs. The training accuracy is 40% and the validation accuracy is 38%. For each of the following suggestions, explain what effect each of them is likely to have on the training and validation accuracies and why, if separately implemented in the next iteration of the training.

- Training for 500 epochs with the same dataset.
- Using early stopping.
- Using 10 hidden layers.

Question 4: Features in a DNN **[6 marks]**

In the paper "Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations", Honglak Lee, Roger Grosse, Rajesh Ranganath, and Andrew Ng show the features learned at different layers of a CNN trained to generate elephant images. You can see some of these features in the following table:

Group of features	Visualisation
A	
B	
C	

For each group, explain in what part of the network you think neurons are most likely to learn these features, and why.

Part 2: Transfer Learning, Generative AI and Multimodal AI [53 marks]

Question 5: Transfer Learning [18 marks]

5.1 Please use 1-2 sentences to explain what is transfer learning. **[4 marks]**

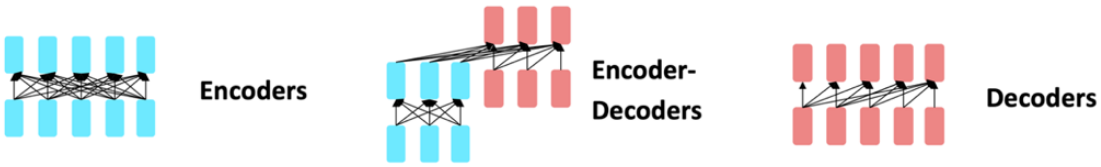
5.2 Please briefly explain what the pre-training and fine-tuning method is. **[4 marks]**

5.3 Which of the following is not a common method in transfer learning? **[2 marks]**

A. Fine-tuning
B. Feature extraction
C. Random initialization
D. Domain adaptation

5.4 Pretrained NLP models typically adopt one of three architectures: encoders, encoder-decoders, or decoders (as shown in the following figure). **[8 marks]**

1) (2 marks) Which architecture does BERT fall under?
2) (4 marks) Please briefly introduce two pre-training tasks utilized in BERT.
3) (2 marks) BERT employs several special tokens during tokenization. Briefly explain the function of [CLS].



Question 6: Generative AI [19 marks]

6.1 Please **choose the most correct answer**: (2 marks per question) **[10 marks]**

1) Autoencoders (AE) typically consist of which two main parts?
   A. Generator and Discriminator
   B. Encoder and Decoder
   C. Encoder and Generator

2) In a Variational Autoencoder (VAE), what is the purpose of the Kullback-Leibler divergence term in the loss function?
   A. To ensure the encoded representations are sparse.

B. To encourage the learned distribution to be close to a prior distribution, like a standard normal.
C. To maximize the difference between the encoder and decoder outputs.

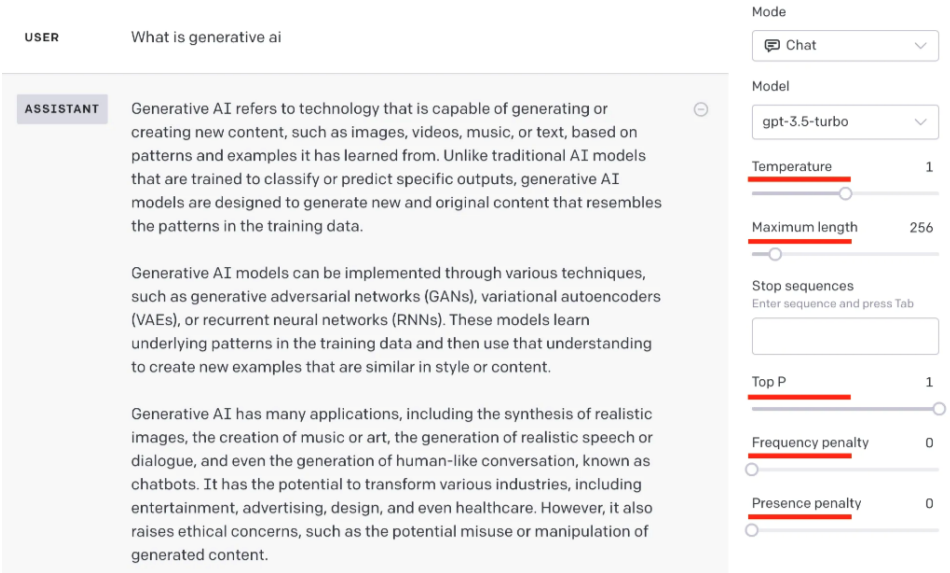
3) Generative Adversarial Network (GAN) typically consist of which two main parts?
   A. Generator and Discriminator
   B. Encoder and Decoder
   C. Encoder and Generator

4) What does the Discriminator in a GAN aim to do during training?
   A. Create realistic images
   B. Tell real data apart from fake data
   C. Increase the noise in data

5) Why do diffusion models use a series of small denoising steps instead of one big jump to clean the data?
   A. It's faster computationally
   B. Easier to learn and control the process
   C. Requires less data

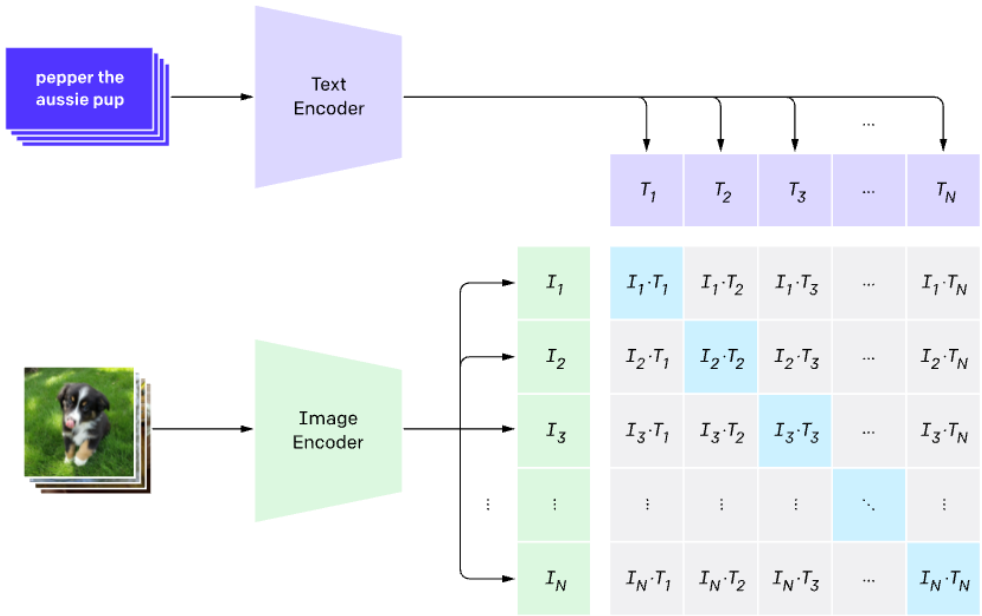
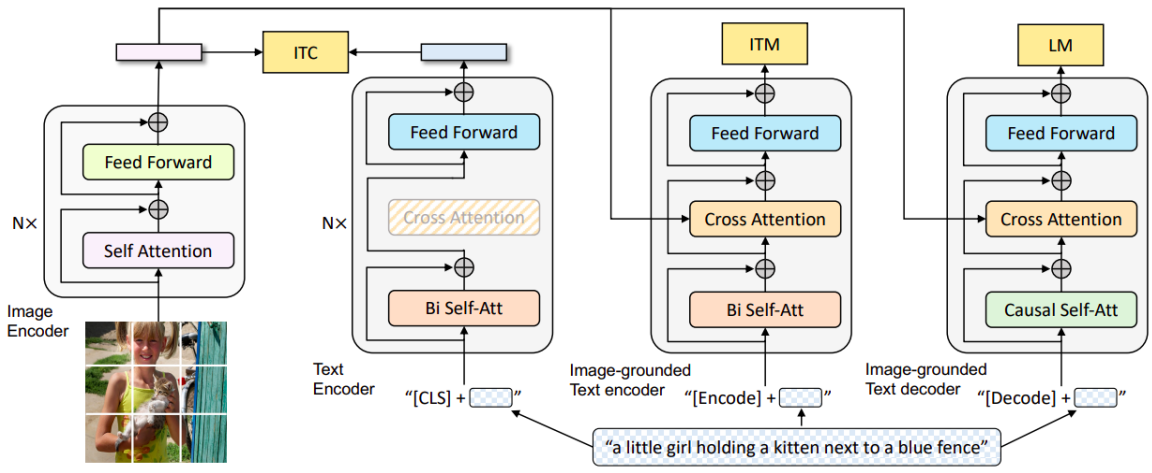
6.2 Please briefly introduce what is a **language model**. **[3 marks]**

6.3 Prompt engineering is important in harnessing the power of GPT. There are key several parameters in prompt engineering (as outlined in the figure on the right). Please explain briefly what the Temperature and Top P parameters are, and their impact on the output with higher values. **[6 marks]**



Question 7: Multimodal AI **[16 marks]**

7.1 The following figure shows the architecture of CLIP. Please briefly explain the infoNCE loss used in CLIP model, including 1) how it encourages discriminative learning between positive (matching) and negative (non-matching) pairs; and 2) how this process aligns the image/text embeddings. **[6 marks]**



7.2 The following figure shows the pre-training model architecture and objectives of BLIP. Please explain the primary objectives of **Image-Text Contrastive (ITC)**, **Image-Text Matching (ITM)**, and **Language Modeling (LM)** components, and **how they collectively contribute to the model's functionality**. **[10 marks]**